DS	**Session**	Tree 1	**Question Information**	Level 1 Challenge 68

Question description

A company has n employees with certain salaries. Your task is to keep track of the salaries and process queries.

Constraints

- $1 \le n,q \le 2 \cdot 10^5$
- $1 \le pi \le 10^9$
- $1 \le k \le n$
- $1 \le x \le 10^9$
- $1 \le a \le b \le 10^9$

Input

The first input line contains two integers n and q: the number of employees and queries. The employees are numbered 1,2,…,n.
The next line has n integers p1,p2,…,pn: each employee's salary.
After this, there are q lines describing the queries. Each line has one of the following forms:

- ! k x: change the salary of employee k to x
- ? a b: count the number of employees whose salary is between a…b

Output

Print the answer to each ? query.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 3
3 7 2 2 5
? 2 3
! 3 6
? 2 3
```

EXPECTED OUTPUT

```
3
2
```

Test Case 2

INPUT (STDIN)

```
5 3
31 17 12 12 15
? 12 31
! 13 6
? 2 13
```

EXPECTED OUTPUT

```
5
3
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int compare(const void
*a,const void *b)
```

Test Case 2

KEYWORD

```
void update(int i,int
n,int x)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
1
```

Test Case 2

TOKEN COUNT

```
575
```

Test Case 3

NLOC

```
65
```

Code

solution.c

```c
#include <stdio.h>

int salary[200005];

int main() {
    int n, q;
    if (scanf("%d %d", &n, &q) != 2) return 0;
    for (int i = 1; i <= n; i++) scanf("%d", &salary[i]);
    while (q--) {
        char type;
        scanf(" %c", &type);
        if (type == '!') {
            int k, x;
            scanf("%d %d", &k, &x);
            salary[k] = x;
        } else if (type == '?') {
            int a, b;
            scanf("%d %d", &a, &b);
            int count = 0;
            for (int i = 1; i <= n; i++) {
                if (salary[i] >= a && salary[i] <= b) count++;
            }
            printf("%d\n", count);
        }
    }
    return 0;
}
```

Output

```
Problems  Output  Debug Console  Terminal  Ports        powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
5 3
3 7 2 2 5
? 2 3
! 3 6
? 2 3
3
2
PS C:\C Programming>
```